Nptel Online Certification Course: Computer Vision
Assignment 1: Fundamentals of Image processing
Number of questions: 10 **Total marks: 10 X 2 = 20**

QUESTION 1: **Type:MCQ**

The probability of a normal variable $P(z > 1.8) = 0.0359$. A random variable X is normally distributed with mean $\mu = 50$ and standard deviation $\sigma = 10$. Compute $P(X < 68)$.

a) 0.9641

b) 0.0359

c) 0.1800

d) 0.8200

Correct Answer: a)
Detailed Solution:
For $X = 68$, the z-value is:

$$z = \frac{X - \mu}{\sigma} = \frac{68 - 50}{10} = 1.8$$

$$P(X < 68) = P(z < 1.8)$$

$$P(z > 1.8) = 0.0359$$

$$P(z < 1.8) = 1 - 0.0359 = 0.9641$$

QUESTION 2: **Type:MCQ**

Consider the following 3-bit grayscale image:

$$I = \begin{bmatrix} 0 & 1 & 2 & 3 \\ 3 & 4 & 5 & 6 \\ 0 & 1 & 2 & 4 \\ 5 & 2 & 4 & 6 \end{bmatrix}$$

What is the optimal threshold according to Otsu method?

a) 1

b) 4

c) 5

d) 2

Correct Answer: d)

Detailed Solution: The histogram is divided into two classes at each possible threshold, and the between-class variance is to be computed.

The histogram of the 3-bit image is:

Gray level (i)	0	1	2	3	4	5	6
$h(i)$	2	2	3	2	3	2	2

Total number of pixels:

$$N = 16$$

Normalized probabilities:

$$p(i) = \frac{h(i)}{16}$$

Using Otsu's method, compute the between-class variance for each threshold.

t	ω_0	ω_1	μ_0	μ_1	σ_B^2
0	0.125	0.875	0	3.57	1.39
1	0.25	0.75	0.50	4.00	2.30
2	0.4375	0.5625	1.14	4.67	**3.06**
3	0.5625	0.4375	1.56	5.00	2.91
4	0.75	0.25	2.33	5.50	1.88
5	0.875	0.125	2.71	6.00	1.18

Since the maximum between-class variance occurs at $t = 2$.

QUESTION 3: **Type:MCQ**

Consider two images each of size 40×40, namely I_1 and I_2. Image I_1 has 900 foreground pixels and 700 background pixels. Image I_2 has 600 foreground pixels and 1000 background pixels. One pixel is selected at random from one of the two images, and it is found to be a background pixel. Find the probability that it was drawn from image I_1.

a) $\frac{17}{7}$

b) $\frac{10}{17}$

c) $\frac{7}{10}$

d) $\frac{7}{17}$

Correct Answer: d)

Detailed Solution:

Let E_1 and E_2 be the events of choosing images I_1 and I_2, respectively. Let A be the event that the selected pixel is a background pixel. Since one image is chosen randomly:

$$P(E_1) = P(E_2) = 0.5$$

For image I_1:

$$P(A|E_1) = \frac{700}{1600}$$

For image I_2:

$$P(A|E_2) = \frac{1000}{1600}$$

Using Bayes' Theorem:

$$P(E_1|A) = \frac{P(A|E_1) \cdot P(E_1)}{P(A|E_1) \cdot P(E_1) + P(A|E_2) \cdot P(E_2)}$$

Substituting:

$$P(E_1|A) = \frac{\left(\frac{700}{1600}\right)\left(\frac{1}{2}\right)}{\left(\frac{700}{1600}\right)\left(\frac{1}{2}\right) + \left(\frac{1000}{1600}\right)\left(\frac{1}{2}\right)}$$

Canceling common terms:

$$P(E_1|A) = \frac{700}{700 + 1000}$$

$$= \frac{700}{1700}$$

$$= \frac{7}{17}$$

QUESTION 4: **Type:MCQ**

Consider the image patch:

$$I = \begin{bmatrix} 3 & 3 & 3 \\ 0 & 0 & 0 \\ -3 & -3 & -3 \end{bmatrix}$$

Which operator is most closely represented?

a) Median filter

b) Histogram equalization

c) Prewitt operator

d) Gaussian smoothing

Correct Answer: c)
Detailed Solution: Prewitt masks use uniform weights to estimate gradients and detect edges in specific directions.

QUESTION 5: **Type:MCQ**

Consider the image patch:

$$I = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$

Which kernel does it correspond to?

a) Averaging filter

b) Sobel operator

c) Laplacian operator

d) Median filter

Correct Answer: b)
Detailed Solution: This is the Sobel mask used for vertical edge detection by emphasizing intensity changes along the y-direction.

QUESTION 6: **Type:MCQ**

Which of the following is the primary purpose of histogram equalization?

a) Reduce image size

b) Improve image contrast

c) Remove image noise

d) Detect edges

Correct Answer: b)
Detailed Solution: Histogram equalization redistributes intensity values over the available range to enhance contrast, especially in low-contrast images.

QUESTION 7: **Type: MCQ**

Consider the following 4×4 image array:

$$I = \begin{bmatrix} 0 & 1 & 2 & 4 \\ 2 & 3 & 4 & 5 \\ 1 & 2 & 4^* & 6 \\ 0 & 3 & 5 & 6 \end{bmatrix}$$

What is the value after histogram equalization at location $(2, 2)$ shown with asterisk mark in the image array?

a) 3

b) 4

c) 5

d) 6

Correct Answer: c)
Detailed Solution:
Histogram table:

Pixel	Frequency	Cumulative Sum	Normalized Sum	New Pixel
0	2	2	0.875	1
1	2	4	1.75	2
2	3	7	3.06	3
3	2	9	3.94	4
4	3	12	5.25	5
5	2	14	6.13	6
6	2	16	7.00	7

The marked pixel value is 4, which maps to 5 after histogram equalization. Hence, the answer is **5**.

QUESTION 8: **Type:MCQ**

Consider the following kernel:

$$K = \begin{bmatrix} 2 & 4 & 6 \\ 1 & 2 & 3 \\ -4 & -8 & -12 \end{bmatrix}$$

What is the rank of this matrix, and is it separable?

a) Rank = 1, separable

b) Rank = 2, separable

c) Rank = 2, not separable

d) Rank = 3, not separable

Correct Answer: a)
Detailed Solution:
Each row is a scalar multiple of the second row:

$$R_1 = 2R_2, \quad R_3 = -4R_2$$

Thus, only one independent row exists, so the rank is 1. A rank-1 kernel can be written as an outer product of two vectors. Hence, the kernel is separable.

QUESTION 9: **Type: MCQ**

Consider a binary image with a fully filled white square at the center on a black background. How does this image look after applying image gradient operation?

a) Entire square remains white

b) A thin line surrounding the square boundary

c) Only the center pixel becomes white

d) Entire background becomes white

Correct Answer: b)
Detailed Solution: The gradient highlights intensity transitions only at object boundaries. Since the square has constant intensity inside, only its edges are detected.

QUESTION 10: **Type: MCQ**

Find the Fourier Transform of the time signal

$$x(t) = e^{-3t}u(t)$$

a) $\frac{1}{3+j\omega}$

b) $\frac{1}{2+j\omega}$

c) $\frac{1}{1+j\omega}$

d) $\frac{1}{4+j\omega}$

Correct Answer: a)

Detailed Solution: Using:

$$X(j\omega) = \int_0^\infty e^{-3t}e^{-j\omega t}dt$$

$$= \int_0^\infty e^{-(3+j\omega)t}dt$$

$$= \frac{1}{3 + j\omega}$$
